

01

Electric Vehicle
Codec Technologies India
Internship 2026

Battery Management System Simulation

96s2p NMC pack model, EKF SOC, SOH ageing, thermal manager, safety FSM (Python)

AT A GLANCE

0.63 %

EKF SOC RMSE (after settling)

-6.64 %

Coulomb-count drift, 30 min

34.0 / 50.9 degC

1.5C fast-charge peak, cooled / not

90.1 % cap.

SOH after 500 cycles

+1.2 %

LS capacity estimate error

7.5 h

Balancing 52 -> 5 mV

5 / 5

Fault flags exercised

23 / 23

Tests re-run here

VERDICT

TESTS 23/23

RE-RUN HERE

DECLARED SYNTHETIC

NO HARDWARE

DETERMINISTIC

Eswar Mahalingam
Electric Vehicle Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
github.com/eswarmahalingam

1. Objective and brief mapping

The brief asked for a simulation of the core functions of an EV battery management system: state-of-charge and state-of-health estimation, thermal management and safety. I implemented all of it as a pure-Python package (**bms/**) with a simulated plant (96s2p pack of 2-RC cells, lumped thermal model, vehicle model) and a BMS layer (Coulomb counter, Extended Kalman Filter, SOH estimator, hysteresis + PI thermal manager, latching fault monitor, passive balancer). **Every number in this report is synthetic**: it comes from the models in this folder, not from a measured cell or vehicle, and is reproduced exactly on every run because all seeds are fixed.

python main.py runs seven closed-loop scenarios (drive, CC-CV charge, fast charge cooled/uncooled, cold start, 500-cycle ageing, balancing, fault injection) in 9.7 s and writes **results/summary.json**, from which every figure below is copied.

2. Model and architecture

Cell: $V = OCV(SOC) - i R_0(T) - v_1 - v_2$ with $\tau_{u1} = 20$ s, $\tau_{u2} = 100$ s and an Arrhenius $R_0(T)$ that doubles from 25 to 0 degC; 13-point NMC-like OCV table (3.00 V at 0 %, 4.20 V at 100 %). Pack: 96 series groups x 2 parallel, 30 Ah cells, 356 V nominal, 21.4 kWh, 1.5 % capacity and 4 % resistance spread (seed 42), 0.25 mOhm/group busbar. The EKF estimates $x = [SOC, v_1, v_2]$ on the average cell with the Joseph-form covariance update and does *not* know the true capacity, busbar drop or the 2 A current-sensor bias.

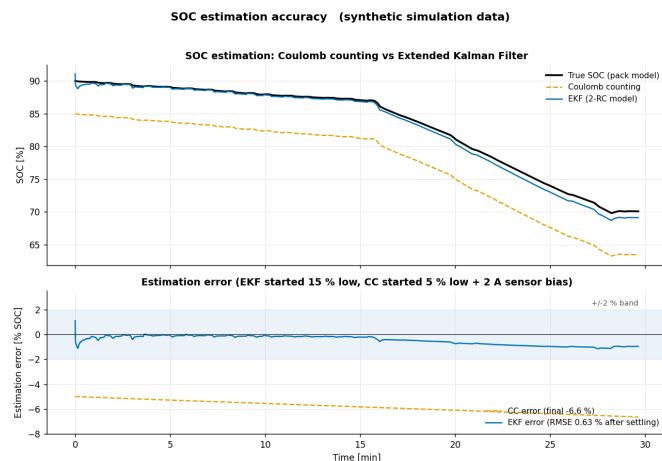


Fig. 1 - True vs Coulomb-counting vs EKF SOC on the 29.6 min synthetic drive; the EKF starts 15 % low and converges in under a minute, the Coulomb counter drifts with the 2 A bias.

3. Method and implementation

- Thermal: $m \cdot c_p \cdot dT/dt = Q_{gen} + P_{heater} - UA_{cool}(flow)(T - T_{cool}) - UA_{amb}(T - T_{amb})$, $Q_{gen} = I^2 R + \text{entropic heat}$; pump ON above 33 degC / OFF below 29 degC, PI flow control (max 12 L/min) towards 30 degC; 2.5 kW heater below 5 degC.
- SOH: Ah-throughput ageing law with Arrhenius temperature dependence and stress(DoD, C-rate); $SOH_{capacity} = Q/Q_{nom}$, $SOH_{resistance}$ from R_0 growth towards 2 R_0 ; least-squares capacity fit of $Ah_{in}(t) = Q \cdot SOC_{EKF}(t) + c$ over the 35-75 % window of a 0.5C charge.
- Safety: OV / UV / OT / UT / OC / IMBALANCE flags with latching, event log and contactor command; passive balancer with 200 mA bleed resistors, inhibited above 5 A pack current.
- Drive cycle: 15 min synthetic stop-and-go (25-50 km/h) + highway (95-120 km/h), $F = ma + \text{aero} + \text{rolling}$, regen capped at 40 kW; not a standard cycle.

4. Results (results/summary.json)

Metric	Value	Comment
Drive: distance / energy / consumption	29.24 km / 4.25 kWh / 145.4 Wh/km	SOC 90.0 -> 70.1 %, peak 157 A discharge, -32.7 A regen
EKF SOC RMSE after settling / full run / during charge	0.63 % / 0.59 % / 0.97 %	max abs error 1.15 %, final -0.96 % (busbar mismatch)
Coulomb counting RMSE / final drift	5.84 % / -6.64 %	5 % initial error + 2 A sensor bias
Drive peak temperature managed / unmanaged	35.9 / 40.9 degC	40 degC ambient, mean coolant flow 2.04 L/min
1.5C fast charge peak cooled / uncooled	34.0 / 50.9 degC	45 degC limit exceeded only without cooling (36 min)
Cold start -8 degC	heater 15.7 min, inhibit 6.8 min	0.654 kWh heater energy, 9.1 degC after 25 min
SOH after 500 cycles (80 % DoD, 35 degC, 1C)	90.1 % capacity / 90.6 % resistance	27.03 Ah left, R_0 +9.4 %
LS capacity estimate	30.35 Ah vs 29.99 Ah true (+1.2 %)	$R^2 = 0.99995$, CC-CV charge 98.4 min

01

Electric Vehicle
Codec Technologies India
Internship 2026

Battery Management System Simulation

96s2p NMC pack model, EKF SOC, SOH ageing, thermal manager, safety FSM (Python)

AT A GLANCE

0.63 %

EKF SOC RMSE (after settling)

-6.64 %

Coulomb-count drift, 30 min

34.0 / 50.9 degC

1.5C fast-charge peak, cooled / not

90.1 % cap.

SOH after 500 cycles

+1.2 %

LS capacity estimate error

7.5 h

Balancing 52 -> 5 mV

5 / 5

Fault flags exercised

23 / 23

Tests re-run here

VERDICT

TESTS 23/23

RE-RUN HERE

DECLARED SYNTHETIC

NO HARDWARE

DETERMINISTIC

Eswar Mahalingam
Electric Vehicle Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
github.com/eswarmahalingam

Metric	Value	Comment
Passive balancing, 192 cells	52.0 -> 5.0 mV, 5.56 -> 0.56 % SOC	flag cleared after 1.74 h, 499 Wh bled
Fault events normal / fault-injection	2 / 6	OV, OC, IMBALANCE, OT, UV raised; contactor opened

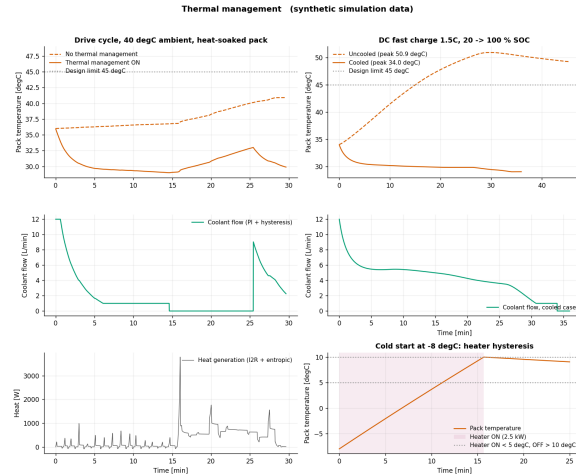


Fig. 2 - Thermal scenarios: drive at 40 degC ambient with and without management, 1.5C fast charge cooled vs uncooled, cold-start heater hysteresis.

4b. Fault-event log (results/summary.json, safety.events)

Two of the ten events occur in normal operation and both are balancing requests near full charge, which is the intended behaviour; the six hard faults appear only in the fault-injection scenario, which deliberately keeps abusing the pack after the contactor opens so that every flag can be exercised in one run (a real BMS would cut current at the first latched fault).

Scenario	t (s)	Flag	Trigger value	Threshold / comment
charge (CC-CV)	4626	IMBALANCE	40.005 mV	40 mV spread flag near 100 % SOC; balancing request, no hard fault
fast_charge_cooled	1653	IMBALANCE	40.002 mV	same mechanism at end of charge
fast_charge_uncooled	1639	IMBALANCE	40.077 mV	same mechanism
cold_start	0	UNDER_TEMP	-7.98 degC	charge inhibited until heater lifts pack above 5 degC (6.8 min)
fault_injection	106	OVER_VOLTAGE	4.2529 V	cell OV limit 4.25 V; latched, contactor opened
fault_injection	500	OVER_CURRENT	320.0 A	abuse current step
fault_injection	500	IMBALANCE	40.923 mV	spread driven by the current step
fault_injection	837	OVER_TEMP	55.005 degC	pack OT limit 55 degC
fault_injection	1579	IMBALANCE	40.005 mV	second request while abuse continues
fault_injection	1731	UNDER_VOLTAGE	2.7962 V	cell UV limit 2.8 V; every hard flag exercised in one run

4c. Ageing figure

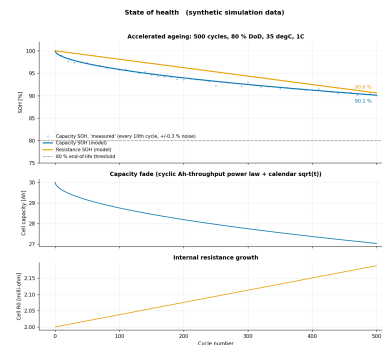


Fig. 3 - 500 accelerated cycles at 80 % DoD, 35 degC, 1C: capacity SOH falls to 90.1 % (27.03 Ah) and resistance SOH to 90.6 % as R0 grows 9.4 %; both curves are monotone (tested). The 80 % end-of-life point is not reached within 500 cycles (cycles_to_80pct_soh = null).

5. Verification

pytest is not installed in this environment, so I re-ran the 23 pytest-style tests through a small runner that provides fixture/approx/parametrize shims and calls every test_* function (2.7 s). **23 passed, 0 failed.**

01

Electric Vehicle
Codec Technologies India
Internship 2026

Battery Management System Simulation

96s2p NMC pack model, EKF SOC, SOH ageing, thermal manager, safety FSM (Python)

AT A GLANCE

0.63 %

EKF SOC RMSE (after settling)

-6.64 %

Coulomb-count drift, 30 min

34.0 / 50.9 degC

1.5C fast-charge peak, cooled / not

90.1 % cap.

SOH after 500 cycles

+1.2 %

LS capacity estimate error

7.5 h

Balancing 52 -> 5 mV

5 / 5

Fault flags exercised

23 / 23

Tests re-run here

VERDICT

TESTS 23/23

RE-RUN HERE

DECLARED SYNTHETIC

NO HARDWARE

DETERMINISTIC

Eswar Mahalingam

Electric Vehicle Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
github.com/eswarmahalingam

Test group	Tests	What is checked	Result
Cell / pack	test_ocv_table_monotonic, test_r0_increases_when_cold, test_cell_disc_harge_lowers_soc_and_voltage, test_pack_aggregation_and_reproducibility	OCV table strictly increasing, inverse lookup exact; R0 doubles when cold; 96 x OCV pack voltage; seed reproducibility	4 PASS
SOC estimation	test_ekf_rmse_below_2pct, test_ekf_converges_from_wrong_initial_soc, test_coulomb_counter_integrates	EKF RMSE < 2 %, max error < 3 %, beats Coulomb counting; convergence from -15 %	3 PASS
Thermal	test_thermal_limit_respected_only_with_management, test_drive_temperature_lower_with_management, test_heater_engages_below_5c, test_thermal_manager_hysteresis, test_thermal_energy_balance_cools_towards_coolant	45 degC limit met only with cooling; hysteresis band; heater logic; energy balance	5 PASS
SOH / ageing	test_soh_monotonic_non_increasing, test_ageing_is_faster_when_hot, test_least_squares_capacity_estimate_within_3pct, test_least_squares_recovers_exact_capacity_on_ideal_data, test_cycle_counting	monotone SOH, Arrhenius ordering, LS error < 3 % and R ² > 0.999, half-cycle counter	5 PASS
Safety / balancing	test_safety_flags_raised_and_latched, test_fault_injection_raises_every_hard_fault, test_no_faults_during_normal_drive, test_passive_balancing_reduces_spread, test_balancer_only_at_rest	every flag latches and opens contactor; spread reduced > 4x; balancer inhibited under load	5 PASS
Drive cycle	test_drive_cycle_is_physical	speeds, accelerations and currents within physical bounds	1 PASS

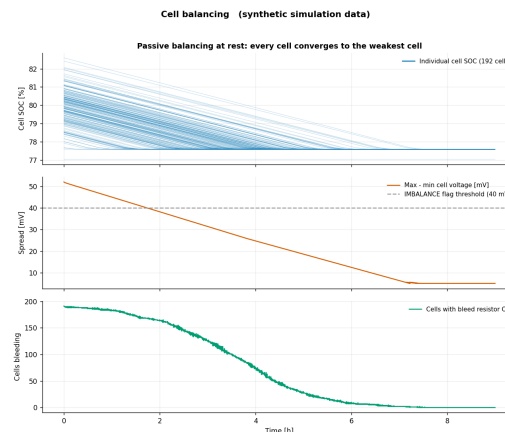


Fig. 4 - Passive balancing of 192 cells with 200 mA bleed resistors: voltage spread 52.0 -> 5.0 mV and SOC spread 5.56 -> 0.56 % in about 7.5 h; the IMBALANCE flag (40 mV) clears after 1.74 h and 499 Wh are dissipated.

6. Data integrity and declared substitutions

- **All data is synthetic.** From the README: "Every voltage, current, temperature and ageing curve is produced by the models in the bms/ package. No real cell, vehicle or fleet measurements were used. Parameter values are chosen to be representative of an NMC/graphite cell and a compact city EV, not to reproduce any specific product."
- **No hardware, no MATLAB/Simulink.** The plant and BMS run in Python only. The README lists Simulink/Simscape Battery + Stateflow as the natural port and Embedded Coder for the estimator; that port was not done here and nothing is claimed to have run in Simulink.
- The drive cycle is synthetic, not WLTC / IDC / FTP-75. The thermal model is lumped (one pack temperature); parallel-cell current sharing is capacity-weighted rather than Kirchhoff-solved; the EKF runs on the average cell, not per cell.

7. Reproducing the numbers

cd 01_BMS_Simulation; python main.py (9.7 s, seeds fixed) regenerates every PNG in results/ plus summary.json, summary.csv and drive_timeseries.csv; python -m pytest (or the shim runner used here) runs the 23 tests in about 3 s. Because every scenario is seeded, the values in this report are reproduced to the printed precision on every run.

8. Learnings

The voltage feedback in the EKF removes both the initial-SOC error and the current-sensor bias; its residual ~1 % error at high current is model mismatch (the estimator does not know the 24 mOhm of busbars), which is exactly what a real BMS calibrates out. A 30 min mixed drive does not overheat a 21 kWh pack even at 40 degC ambient, but 1.5C fast charging without coolant passes 45 degC after about 14 min - the thermal design case is the charger, not the road. Passive balancing is slow (hours) and dissipative (0.5 kWh here), so it only makes sense at rest or at the end of charge. Next steps I would take: per-cell (bar-delta) EKF, Kirchhoff-solved current sharing, a rainflow cycle counter and standard drive cycles from published tables.